

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 10)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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6	
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8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

- A Ozone is destroyed only by CFCs ☐
- B Ozone absorbs mainly infrared radiation ☐
- C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐
- D Ozone concentrations are highest at ground level ☐

0 1 . 2

Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 1 . 3

Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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5

0 2

A motor-racing circuit has residents living 500 m away. The circuit operator wants to compare two designs of acoustic barrier: an earth baffle mound and a timber acoustic fence.

0 2 . 1

Describe how you would carry out an investigation to compare the effectiveness of the two barriers.

[5 marks]

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0 2 . 2 Explain why readings should be taken with the same wind direction. **[1 mark]**

0 2 . 3 Explain the difference between the LA10 and LA90 measures of traffic noise. **[2 marks]**

0 2 . 4 Outline **two** ways in which the residents' houses could be designed to reduce the noise reaching occupants. **[2 marks]**

1

2

10

03 . 1

Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

[3 marks]

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03 . 2

Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

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03 . 3

Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

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03 . 4

Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

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10

0	4	A coal-mine spoil heap on a hillside has been abandoned. The spoil contains shale with some iron pyrites and residual coal.
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0	4
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1

 Explain **three** problems this spoil heap could cause and describe how each could be managed.

[6 marks]

04 . 2 Explain why deep layers of soil are applied when a spoil heap containing toxic materials is restored.

[2 marks]

0 4 . 3

Explain why an old steel-works spoil heap may be valuable for wildlife conservation.

[2 marks]

10

0 5

Table 1 compares the energy used to produce 1 tonne of aluminium from bauxite ore with the energy used to produce 1 tonne from recycled aluminium cans.

Table 1

Source	Energy used / GJ per tonne	CO ₂ emitted / tonnes per tonne of aluminium
Bauxite ore (extraction, refining, smelting)	210	12
Recycled cans (collection, sorting, remelting)	12	0.7

0 5 . 1

Calculate the percentage energy saving from recycling aluminium rather than extracting it from ore.

[2 marks]

Answer

%

0 5 . 2

Give **three** advantages of recycling metals other than energy saving.

[3 marks]

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0 5 . 3 Explain **four** difficulties that reduce the effectiveness of recycling schemes. **[4 marks]**

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0 5 . 4 Explain what is meant by **Cradle to Cradle** design. **[1 mark]**

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0 6

Table 1 shows information about some greenhouse gases.

Table 1

Gas	Main anthropogenic sources	Approximate residence time	Global warming potential (per molecule, relative to CO₂ over 100 years)
Carbon dioxide	Fossil fuel combustion, deforestation, cement	100+ years (variable)	1
Methane	Rice paddies, cattle, landfill, gas leaks, coal mines	About 12 years	28
Nitrous oxide (NO _x)	Fertilisers, combustion	About 120 years	265
CFCs	Refrigerants, aerosols (now banned)	50-100+ years	5 000-10 000
Tropospheric ozone	Secondary pollutant from NO _x and hydrocarbons	Days-weeks	-

0 6**. 1**

Explain the natural greenhouse effect.

[3 marks]

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0 6**. 2**

Use Table 1 to explain why carbon dioxide is considered the most important anthropogenic greenhouse gas even though methane and CFCs absorb far more infrared per molecule.

[3 marks]

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0 6 . 3

Explain why reducing methane emissions would produce a faster reduction in warming than reducing CO_2 emissions by the same mass.

[2 marks]

0 6 . 4

Give **two** methods of reducing methane emissions from agriculture or waste.

[2 marks]

1

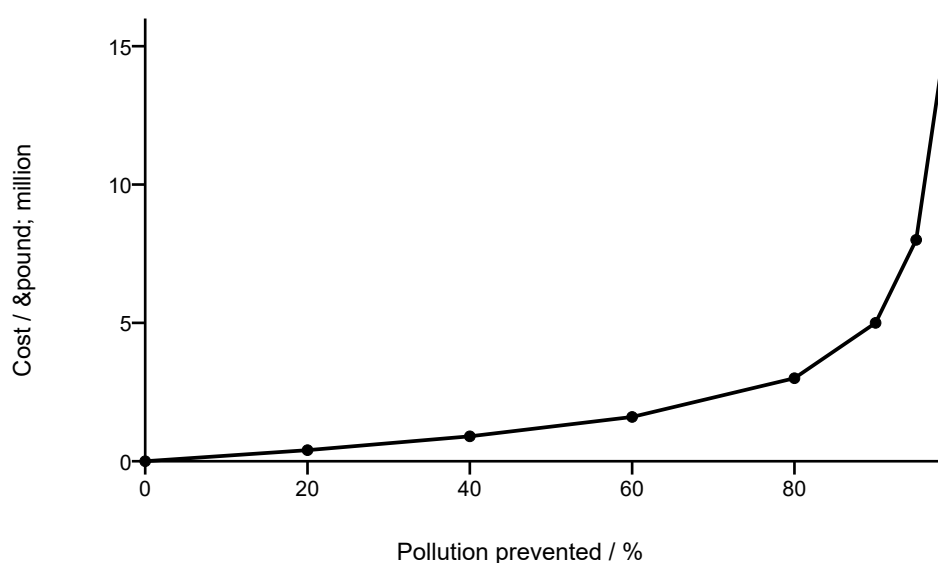
2

10

0 7

Figure 1 shows the relationship between the cost of pollution control equipment at a factory and the percentage of a pollutant emission that is prevented.

Figure 1



07 . 1

Describe the relationship shown in Figure 1.

[2 marks]

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07 . 2

Use Figure 1 to calculate the extra cost of increasing the pollution prevented from 90% to 99%.

[1 mark]

Answer

£ million

07 . 3

Explain what is meant by the terms **ALARA** and **BATNEEC** and how they would be applied by the factory.

[4 marks]

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07 . 4

Suggest why the control methods used originally by a factory may no longer be sufficient after several years.

[3 marks]

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10

08 . 1 Explain **three** advantages of biofuels over other renewable energy resources. [3 marks]

08 . 2 Explain **three** disadvantages of biofuel crops. [3 marks]

08 . 3 Describe **two** new biofuel technologies. [2 marks]

1

2

0 8 . **4** Explain how new low-temperature turbine technology could allow geothermal electricity generation in the UK.

[2 marks]

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10

0 9 In 2020 the tanker MV Wakashio ran aground on a coral reef off Mauritius and leaked about 1000 tonnes of fuel oil into a shallow lagoon near mangroves and a marine park.

0 9 . **1** Explain why inflatable booms were deployed immediately, and give **one** limitation of booms.

[3 marks]

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0 9 . **2** Describe how skimmers and absorbent materials remove oil, and state how the collected oil is disposed of.

[3 marks]

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09 . 3

Explain why dispersants were **not** used in the lagoon.

[2 marks]

09 . 4

Explain how bioremediation could help remove the oil that remained on the mangrove sediments, and suggest how it could be accelerated.

[2 marks]

10

10 . 1

Discuss how an understanding of the properties of pollutants can be used to predict their behaviour in the environment and to plan strategies to control them.

[9 marks]

[illegible]

10 . 2 State what is meant by **bioaccumulation**.

[1 mark]

10

Do not write outside the box

Write an essay on one of the following topics.

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Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

OR

1 1 . 2

Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

Do not write outside the box

END OF QUESTIONS